



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO ESWARI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

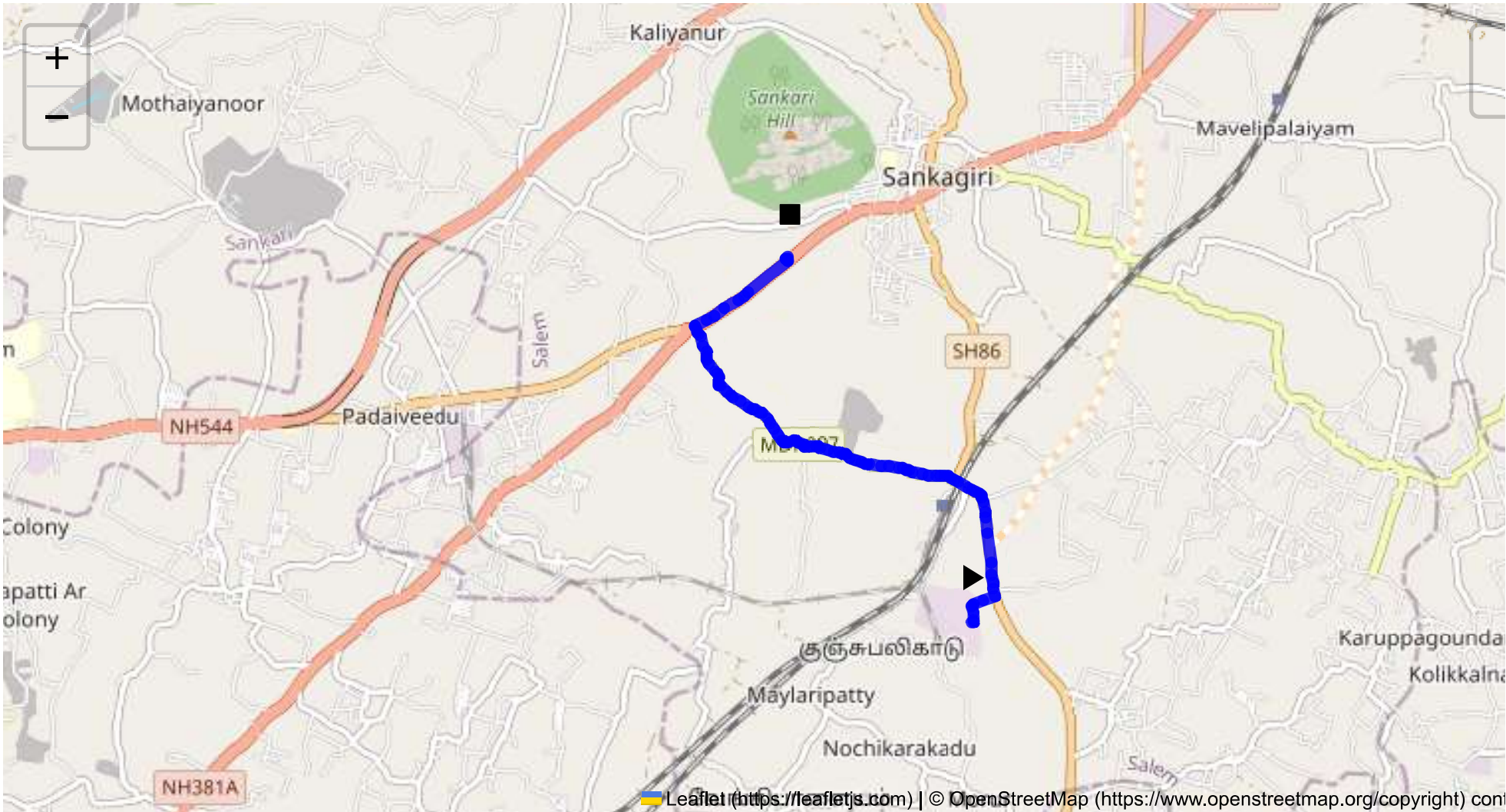
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 5.92 km
Estimated Duration: 0.2 hours
Adjusted Duration (Heavy Vehicle): 0.2 hours
Start: (11.4381, 77.8734)
End: (11.469146, 77.857364)



Welcome to the Journey Risk Management Study

Certainly, here's a comprehensive analysis of the route from CVQF+23W, Sangagiri, Tamil Nadu 637301, India to 3/64, Pallipalayam - Sankari Rd, Sangagiri, Tamil Nadu 637301, India.

1. Overview of the Route Map

This route stretches approximately 5.92 kilometers, mostly consisting of local roads. The journey begins in a semi-urban area and continues through a region that combines residential and commercial land use, typical of small Indian towns transitioning into more urbanized areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Patterns:** The climate is generally tropical, with hot summers and a moderate monsoon season from June to October.
- **Weather-Related Hazards:** During the monsoon season, heavy rain can cause waterlogging and poor visibility. This increases the risk of skidding and impacts road quality.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (7–9 am) and evening (5–7 pm) typically see increased traffic due to local commuting.
- **Congestion-Prone Areas:** The entry and exit points of Sangagiri, particularly near marketplaces and schools, tend to become congested.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Overall, the roads are paved but may have potholes and uneven surfaces due to heavy usage and weather conditions.
- **Infrastructure:** Road signage is moderate, with occasional visibility issues; proper lighting may be lacking in some segments.

5. Suggestions for Alternative Routes for Emergencies

- **Diversion through NH 544:** If the main route is obstructed, using NH 544 offers a viable detour, although it increases travel time.
- **Local Streets:** Smaller streets can be used in case of blockages but may require awareness of their condition and load capacity.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials is subject to permits and specific timing restrictions (often avoiding transport during peak traffic hours). Compliance with the Motor Vehicles Act and hazardous materials guidelines is essential.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Historical Incidents:** There have been few major incidents reported; however, there are occasional reports of minor accidents, often attributed to driver error or skidding on wet roads.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Proximity to residential zones requires adherence to noise and air quality controls. Schools and hospitals necessitate caution due to vulnerable populations.
- **Environmental Zones:** None with significant restrictions along this route, though adherence to general pollution norms is encouraged.

9. Analysis of Communication Coverage

- **Coverage:** Mobile network coverage is generally reliable across the route, but there might be brief dead zones in low-lying or densely constructed areas.

10. Estimated Emergency Response Times for Different Route Segments

- **Indicative Emergency Response:** Urban areas may offer quicker response (15–20 minutes), while more rural or congested segments can see delays up to 30 minutes. Proximity to local health centers and police stations facilitate quicker assistance.

12. Overall Summary of Risk Assessment

Overall, the route presents moderate risk under typical conditions, primarily associated with road conditions and occasional congestion. Weather can exacerbate risks, and adherence to timing and regulatory guidelines will mitigate many potential concerns. Comprehensive preparatory measures, including awareness of alternative routes, are essential for safe transit of hazardous materials.

In conclusion, while the route is generally safe for the transportation of goods, especially hazardous materials, diligent attention to changing weather and traffic conditions, adherence to local regulations, and proactive communication with emergency services are vital for minimizing risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Blind Spot	Blind Spot	11.46330, 77.84929	10 KM/Hr
5	Blind Spot	Blind Spot	11.46897, 77.85738	10 KM/Hr
6	Turn	Medium	11.46910, 77.85726	30 KM/Hr